

2 types of segments



→ 4 problems

2 Pointers Problem Solving

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Problem 1



→ Shortest Subarray with Sum at least K [includes negative elements]:

<https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/description/>

→ It might seem like Type 2,

BUT its not.

- maintain a set of <value, index>
- we want to fix the R index
- we can iterate over all <L-1> indices such that
$$p[r] - p[l-1] \geq k$$
 And
remove set $\langle p[l-1], l-1 \rangle$ from the

TASK

$O(n \log n)$

↳ deque / Queue optimizations

↳ $O(n)$

{ -1, 4 }

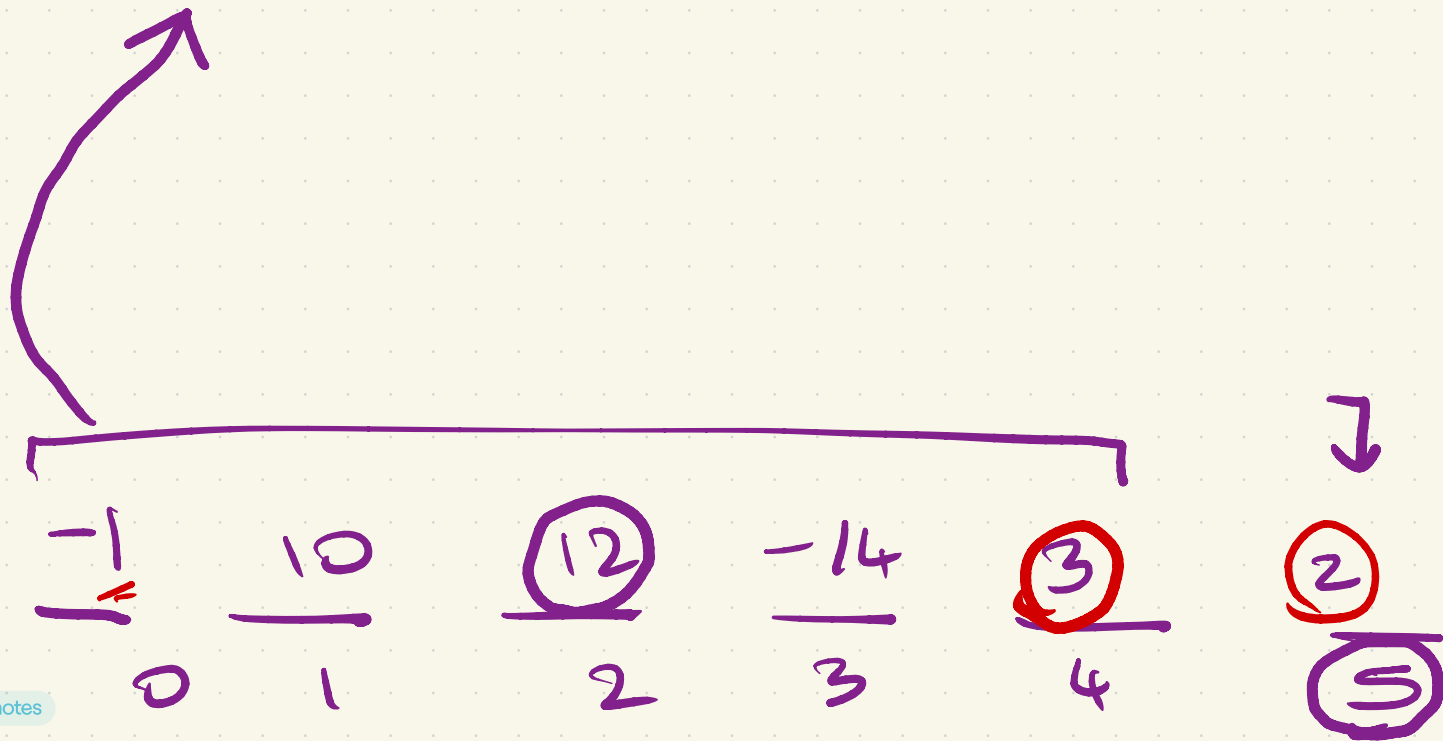
~~-10~~ ~~-2~~ ~~-1~~ ~~0~~ 10 19

↑

4

$k = 10$

{ 3, 0, 5 } deque





```
class Solution {
public:
    int shortestSubarray(std::vector<int>& nums, int k) {
        int n = nums.size();
        deque<int> deque;
        vector<long long> prefixSum(n + 1, 0);

        for (int i = 0; i < n; ++i) {
            prefixSum[i + 1] = prefixSum[i] + nums[i];
        }

        int result = n + 1;
        for (int i = 0; i <= n; ++i) {
            while (!deque.empty() && prefixSum[i] - prefixSum[deque.front()] >= k) {
                result = std::min(result, i - deque.front());
                deque.pop_front();
            }
            while (!deque.empty() && prefixSum[i] <= prefixSum[deque.back()]) {
                deque.pop_back();
            }
            deque.push_back(i);
        }

        return result == n + 1 ? -1 : result;
    }
}
```

$O(n)$ time
 $O(n)$ space

inserting
at

$[R - L + 1]$
 $\rightarrow (R - (L - 1))$

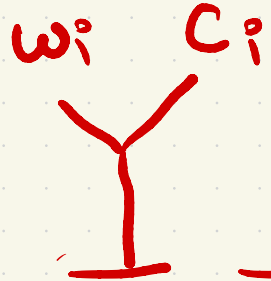
Problem 2



Knapsack on a Segment:

<https://codeforces.com/edu/course/2/lesson/9/3/practice/contest/307094/problem/E>

→ Type 1

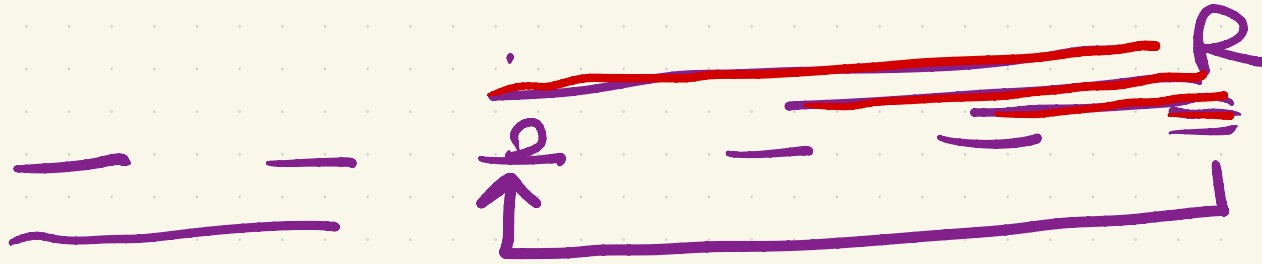


$\sum w_i \leq S$
 $\sum C_i$ is as large as possible

$$w_i, C_i \geq 0$$

$R \rightarrow$ what should be our answer

$\hookrightarrow$ Largest subarray with $\sum w_i \leq S$



$$S = 13$$

$$\begin{array}{cccc|cc} 1 & 2 & 3 & 4 & S & 6 \\ \hline \end{array}$$

→ For every Q , we find the
leftmost L & our final answer
is max of all possible answers

```
vector<int>w(n), c(n);
```

```
for(auto &i : w) cin >> i;  
for(auto &i : c) cin >> i;
```

```
for(int left = 0, right = 0; right < n; right++) {  
    weight += w[right];  
    profit += c[right];
```

adding
a right

```
    while(weight > W) {  
        → weight -= w[left];  
        → profit -= c[left];  
        → left++;  
    }
```

} moving
ahead left 1 step

```
    answer = max(answer, profit);  
}
```

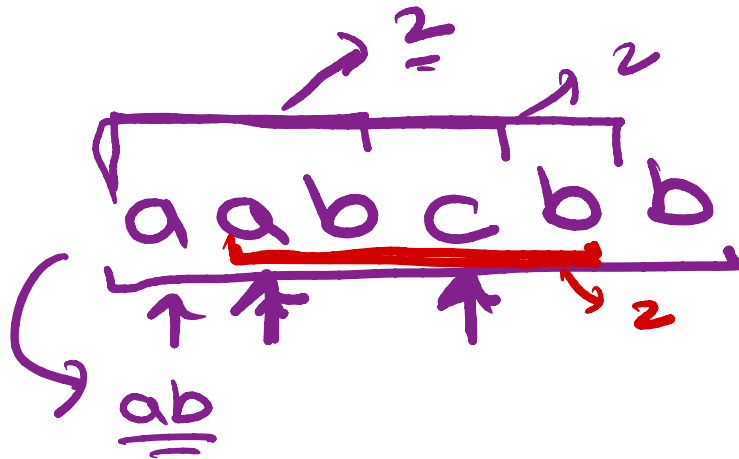
```
cout << answer << "\n";
```


Problem 3



Not Very Rude Substring:

<https://codeforces.com/edu/course/2/lesson/9/3/practice/contest/307094/problem/G>



$$\frac{C = 2}{4}$$

a a b b

Diagram showing a sequence of four characters: a, a, b, b. The first 'a' has a caret (^) above it. The second 'b' is circled. An arrow points up from the circled 'b'. A bracket below the sequence spans from the first 'a' to the second 'b'.

→ 4
=

1 b b a b

a b c d e f g h → 1

Type 1

maximum length substring
with rudeness $\leq C$

number of as

$$3 + \underline{2} = 5$$

+

a b a b b

b rudeness doesn't
change

+ b $\rightarrow$ rudeness + = freq a

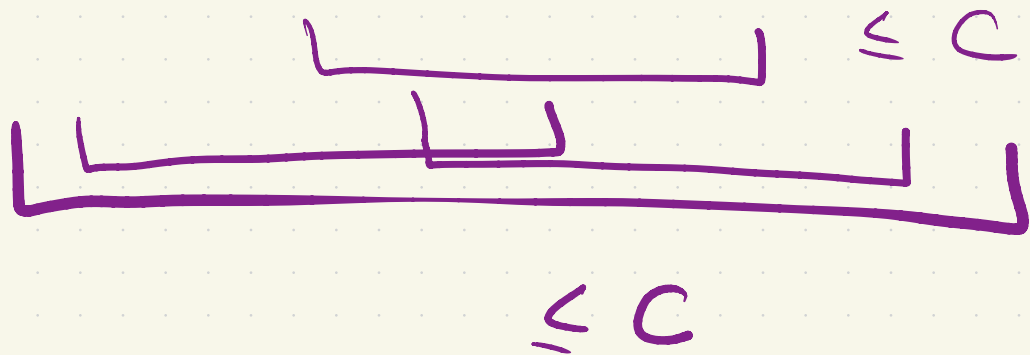
- a $\rightarrow$ rudeness - = freq b

c
a b b a b

$\rightarrow 4$

$\rightarrow 1$

R $\rightarrow$ smallest L
rudeness [L...r]
 $\leq C$



```

for(int left = 0, right = 0; right < n; right++) {
    int a += (s[right] == 'a');
    int b += (s[right] == 'b');
    if(s[right] == 'b') curr += a; ↑

    while(curr > c) {
        → a -= (s[left] == 'a');
        → b -= (s[left] == 'b');
        ↓ if(s[left] == 'a') curr -= b;
        left++;
    }
    curr ≤ c
    answer = max(answer, right - left + 1);
}

cout << answer << "\n";
}

```

Problem 4



Card Substrings:

<https://codeforces.com/edu/course/2/lesson/9/3/practice/contest/307094/problem/F>

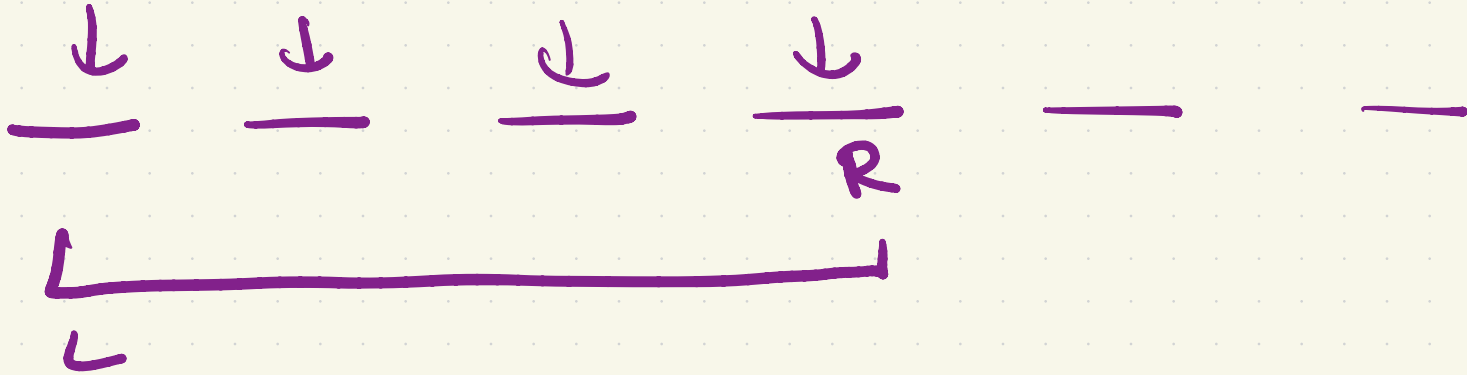
[Count the number of substrings
that we can make in S using
T.]

Type 1



↳ we would fix r and
count how many Ls are
there

$R-L+1$ choices



Check ??

$s = \underline{a} \underline{b} \underline{b} \underline{b}$

$t = \underline{a} \underline{b} \underline{b} \underline{c} \underline{b} \underline{b}$

freq(26)

$i \in [a, z]$
 $freq_i \leq freq_i$

$O(n \cdot 26)$
 $\uparrow$
 $10^8 \times 26$
 2×10^6 ccb8

$$t = \underline{4 \text{ bs}} \quad \text{I}$$

$$s = \underline{5 \text{ bs}} \quad \text{vari}$$

$$\underline{f_{rt_i} - f_{rs_i} \geq 0}$$

$$4 - 3 \geq 0 \quad \text{I}$$

$$\left(\underset{\uparrow}{4} - \underset{\downarrow}{5} = \textcircled{-1} \right)$$

for $(i=0; i < 26; i++)$ {
 $f_{rs_i} \underline{\neq} f_{rt_i}$
 }



vector<int>freq(26);

↓ 1 freq
↓

2 freq
map<char, int>

for(auto &i : t) freq[i - 'a']++;

for(int left = 0, right = 0; right < n; right++) {

→ freq[s[right] - 'a']--; consumed

[while(*min_element(freq.begin(), freq.end()) < 0) {

→ freq[s[left] - 'a']++; ← remove it

→ left++;

}

answer += (right - left + 1);

right - left + 1

cout << answer << "\\n";